PRATHVI SAHU

Software Development Engineer (Full-Stack Java / Spring Boot & React)

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PROFESSIONAL SUMMARY

Computer Science and Design engineering graduate with hands-on experience architecting and shipping full-stack applications using Java, Spring Boot, Python, and React. Built an AI-powered face recognition attendance platform with 50+ secure REST APIs serving 500+ daily active users, and engineered F.R.I.D.A.Y., a voice-controlled AI operating system and real-time trading station featuring ~35ms local system automation. Strong foundation in relational database design, distributed architectures, data structures & algorithms (DSA in Java), Docker containerization, and modern cloud deployment (AWS & Vercel). Seeking an SDE role to build scalable, resilient software products.

TECHNICAL SKILLS

Languages: Java (17 / 21), Python (3.11+), JavaScript (ES6+), SQL, HTML5, CSS3
Frameworks & Libraries: Spring Boot 3, Spring Security, FastAPI, React 19, Vite, JavaFX, OpenCV, Tailwind CSS, Framer Motion
Backend & Distributed Systems: RESTful API Architecture, Microservices, JWT, RBAC, WebSockets, Apache POI, Maven, Redis
AI, Audio & Vision: Groq (Llama 3.3 70B), Google Gemini 2.5, Google MediaPipe (Hands), Web Audio API (DSP Synthesis), Edge-TTS
Databases & Cloud: MySQL, PostgreSQL, SQLite (WAL mode), Supabase, AWS (EC2, S3, Lambda), Docker, Vercel, Git/GitHub, CI/CD
Core Computer Science: Data Structures & Algorithms (DSA), OOP, DBMS, System Design Fundamentals, Operating Systems

KEY PROJECTS

1. F.R.I.D.A.Y. — Voice-Controlled AI Operating System & Real-Time Trading Platform
React 19, FastAPI, Groq Llama 3.3, Gemini 2.5, SQLite, Three.js

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- **Dual-LLM Brain & Neural Voice:** Architected a hybrid neural brain pairing **Groq Llama 3.3 70B (~150ms TTFT)** with Gemini 2.5 failover, real-time **Edge-TTS chunked audio streaming (sub-180ms TTFA)**, and semantic SQLite RAG memory.
- **Fast-Path System Automation:** Built a deterministic intent classification engine **averaging ~35ms end-to-end**, executing **47 system automation tools** across desktop workflows, Spotify, calendar/email, and WhatsApp.
- **Financial Trading Station:** Engineered a live market workstation featuring **TradingView interactive charts**, live OHLCV WebSocket streaming for **5,000+ financial symbols**, and real-time technical indicators (**RSI-14, MACD, EMA-20/50**).
- **AI Career OS & Smart Agents:** Developed an AI-powered career platform with real-time job scraping, ATS resume scoring algorithms, and cryptographically verified approval-first application packet assembly.
- **Modern HUD Architecture:** Designed a 60 FPS sci-fi glassmorphic dashboard using **React 19, Three.js GLSL shaders, and WebSockets** for real-time bidirectional telemetry and voice audio visualization.

2. AI-Powered Face Recognition Attendance Management System (FacetrackU)
Java, Spring Boot 3, Spring Security, SQL, JavaFX, OpenCV, MySQL, JWT

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- Architected and deployed an automated attendance management platform with biometric face recognition, reducing manual marking time by **85% for 500+ daily active users**.
- Configured **Spring Security** with stateless **JWT authentication**, Role-Based Access Control (RBAC), and AES encryption across **50+ RESTful API endpoints**.
- Engineered multi-shot facial enrollment pipelines using OpenCV, maintaining **95%+ verification accuracy** under variable real-world lighting conditions.
- Modeled and optimized a normalized **MySQL relational schema** handling **10,000+ student records** with indexed queries achieving sub-100ms response times.

3. AirChord — AI Virtual Guitar & Gesture-Driven Chord Companion
React 19, TypeScript, Web Audio API, MediaPipe, Karplus-Strong DSP

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- Built a real-time computer vision virtual instrument using **Google MediaPipe (21 3D hand landmarks)**, mapping singer hand gestures to guitar chords with **sub-30ms landmark processing latency**.
- Engineered a physical-modeling audio engine implementing Karplus-Strong string synthesis with acoustic body resonance, pick noise transients, micro-timing jitter (±1.2ms), and per-string stereo panning.
- Implemented an interactive practice room with visual pose guidance, real-time pitch validation, and synchronized video/audio performance recording via MediaRecorder API.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science and Design — **2022 – 2026 (Graduation: Jun 2026)**
New Horizon Institute of Technology and Management, Thane (Mumbai University) — *Thane, India*

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Cloud Computing, Operating Systems, Software Engineering.
- **Problem Solving & DSA:** Active competitive programming and problem-solving practice in Java across Arrays, Strings, HashMaps, Two Pointers, and Recursion.

ACHIEVEMENTS & CERTIFICATIONS

- **Zepto (ZDL) SDE Offer:** Received Software Development Engineer (SDE) offer letter following competitive technical assessments and evaluations in DSA, problem solving, and full-stack development.
- **College Hackathon Participant:** Collaborated with a 4-person team to prototype full-stack software solutions under strict 24-hour sprint deadlines.
- **Engineering Team Lead:** Directed a 4-student software engineering team, coordinating REST API contracts, database schema designs, and sprint milestones.
- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate** — Oracle Corporation
- **AWS Solutions Architecture** — Amazon Web Services (AWS)
- **Deloitte Data Analytics & Cyber Security Virtual Job Simulations** — Tableau & Threat Mitigation